SonoSeeder

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Abstract—**Modern agriculture faces growing pressure from abiotic stressors, persistent yield gaps, and the environmental externalities of intensive chemical inputs, necessitating innovative autonomous solutions. This paper presents SonoSeeder, developed as a scaled-down proof of concept for a multi-functional, solar-powered agro-technological robot that integrates field preparation, precision seeding, irrigation, and ultrasound-assisted physiological enhancement into a single autonomous system. The robot uniquely combines a subsurface thermal ploughing module capable of instantaneous, non-chemical elimination of the weed seed bank via conductive heat transfer, a servo-driven precision plantation mechanism, and a mobile irrigation unit optimized for high water-use efficiency. Crucially, it incorporates an onboard ultrasound chamber experimentally validated to induce acoustic cavitation and sonoporation, thereby accelerating seed germination via α-amylase activation and priming antioxidant defense mechanisms. Built on an ESP32 microcontroller with sensor fusion and PID-based control, SonoSeeder operates fully autonomously. Results from preliminary testing demonstrate superior growth rates in ultrasound-treated seeds, validating the system's biophysical efficacy. By consolidating multiple farming tasks into a single sustainable platform, SonoSeeder offers a cost-effective, scalable solution for crop health and resilience, pesticide and herbicide use, input inefficiencies, and labor shortages in both smallholder and large-scale agricultural systems.**

I. Literature Review

The agricultural sector in Pakistan faces a complex nexus of biophysical and environmental challenges that severely constrain productivity, primarily manifested as persistent yield gaps exceeding 60% for major staples like wheat and rice compared to global best practices [1]. This deficit is largely attributed to unbalanced nutrient management, specifically the over-application of nitrogen (N) and phosphorus (P) coupled with a pronounced neglect of potassium (K); data indicates that N:P:K usage ratios are often skewed to 4:1:0.1, drastically reducing fertilizer use efficiency and stress tolerance [2]. Simultaneously, salinity affects over 6.1 million hectares of arable land (approx. 30% of the irrigated area), driven by rising water tables and the use of groundwater with salt concentrations often exceeding 1000 ppm, which restricts crop physiology and cuts yields by 25–60% depending on crop sensitivity [3], [4]. Beyond soil constraints, erratic climatic patterns induce severe oxidative stress; heat waves alone can reduce cereal yields by 10–30% per degree Celsius rise above optimal temperatures due to the overproduction of Reactive Oxygen Species (ROS) [5], [6]. Furthermore, heavy metal contamination (Cadmium, Lead) in peri-urban soils has reached toxic levels, with transfer factors exceeding 0.5, posing severe health risks [7].

These yield-limiting factors are compounded by a reliance on inefficient traditional technologies [8], while the intensive use of chemical herbicides has led to water contamination, with residue levels in groundwater frequently exceeding World Health Organization (WHO) safety limits of 0.1 μg/L [9]. Additionally, the burning of 8–10 million tons of rice residue annually in Punjab releases massive loads of Particulate Matter (PM2.5) and greenhouse gases, contributing to smog that costs the economy up to 6.5% of GDP annually due to public health crises and lost productivity [10], [11]. This practice also incinerates approximately 80% of the nitrogen and 25% of the phosphorus contained in the straw, creating a negative nutrient feedback loop [12]. To address these issues, recent literature supports the integration of Ultrasound (US) technology and Subsurface Thermal Ploughing. Low-to-medium frequency ultrasound offers a transformative solution by "vaccinating" seeds; research confirms US exposure increases the activity of antioxidant enzymes (SOD, CAT) by 20–45%, effectively neutralizing ROS generated by environmental stress [13], [14]. Furthermore, US treatment induces the accumulation of proline by up to 50%, allowing plants to maintain turgor pressure under severe drought conditions [15].

Ultrasound technology revolutionizes input efficiency by enhancing Nitrogen Use Efficiency (NUE); studies show US treatment can increase leaf nitrogen content by 16–22% and chlorophyll content by 18% even without reduced fertilizer application, suggesting a potential to cut synthetic fertilizer inputs by 20% while maintaining yield [16]. This efficiency stems from a 30–40% increase in lateral root density and sonoporation, which enhances cell membrane permeability for superior ion uptake [17]. Crucially, US-assisted phytoextraction enhances heavy metal removal efficiency by 15–60%, enabling safe cropping in contaminated zones [18]. High-intensity US also accelerates germination by 20–30% (reducing time to emergence by 1–2 days) via acoustic cavitation that activates α-amylase [19], and triggers Systemic Acquired Resistance (SAR), reducing pathogen infection rates by 40–60% without chemicals [20]. Additionally, stress stimulation boosts secondary metabolite production (e.g., flavonoids) by 35–50%, enhancing nutritional value [21].

Complementing this is subsurface instantaneous thermal ploughing, which targets the weed seed bank as a direct substitute for herbicides. The efficacy of this method relies on conduction to raise weed seed temperatures to lethal thresholds (70–80°C) in under 1 second [24]. This achieves a mortality rate of 95–100% for seedlings and up to 80% for dormant seeds in the topsoil [22], [25]. By eliminating the first wave of competition, this method can reduce total herbicide volume by 50–100% depending on the crop system [23]. Unlike chemical removal, thermal weeding leaves killed biomass *in situ*, which decomposes to return approximately 30–50 kg N/ha back to the soil, turning pests into a nutritional resource that directly supports yield gains [26], [27].

II. Introduction

To address critical agricultural challenges, SonoSeeder is an autonomous agro-technological robot designed by LGS Defence Team B from Pakistan for the Future Innovators category at WRO International 2024 in Izmir, Türkiye. The design integrates four subsystems: a subsurface thermal ploughing module utilizing soldering iron heating elements and their metal tips for weed elimination, a servo-driven seeding mechanism, a mobile irrigation sprinkler, and a custom-built ultrasound chamber for physiological seed enhancement. An ESP32 microcontroller paired with a magnetometer and high-torque encoder motors are used for precise navigation and control. The system is powered by an 11.1V 2200mAh LiPo battery, actively recharged by two 6V solar panels mounted atop the chassis in series.

III. Self Validation

To validate the claims surrounding the SonoSeeder's ultrasound chamber, a comparative test was conducted using a prototype version of SonoSeeder's ultrasound subsystem—powered by a repurposed Arduino distance-sensor transducer driven by a 555 timer configured at 40kHz with a 50% duty cycle and supplied by a 9V battery.

This was conducted as a simple qualitative experiment designed to demonstrate that the ultrasound subsystem functions as intended, rather than producing precise quantitative measurements.

Figure 1: Day 1 of the experimental trial.

Figure 2: Day 2 of the experimental trial.

Figure 3: Day 3 of the experimental trial.

Figure 4: Day 4 of the experimental trial.

Figure 5: Day 5 of the experimental trial.

Day	Ctrl (cm)	Exp (cm)	Gain
1	–	–	–
2	–	–	–
3	2.6	4.2	+61.5%
4	4.7	6.5	+38.3%
5	6.8	8.3	+22.1%

Table 1: Maize seedling height comparison between control and ultrasound-treated groups.

IV. Mechanical Construction

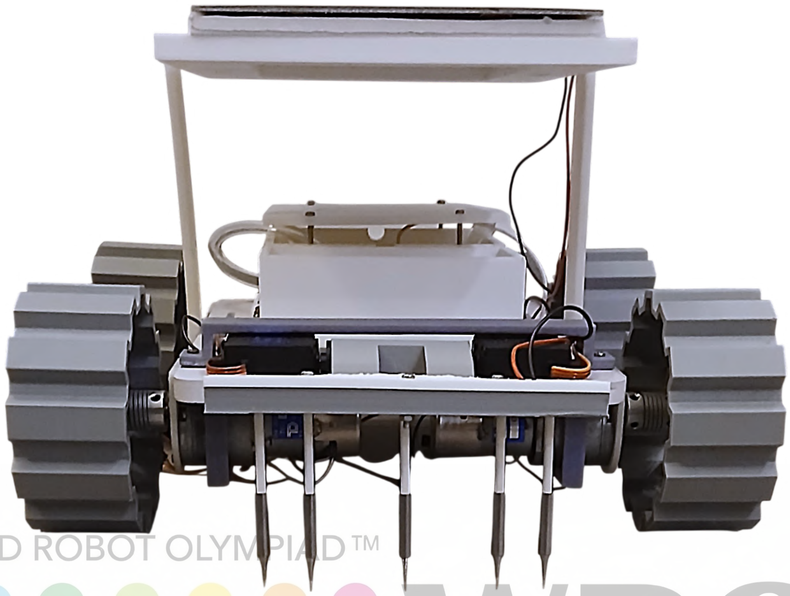

Figure 6: Complete Frontal View of SonoSeeder.

Figure 7: SonoSeeder baseplate, CNC-machined from 1 cm thick PVC (20cm x 30cm). The material's ductility allows for easy manual fastening of components using sharp-tipped screws.

Figure 8: Field preparation compartment assembly. Two servo motors are secured to the front of the baseplate via a custom holder, which is screwed directly into the PVC. The motors control the field preparation compartment.

Figure 9: Servo system attached to the field preparation compartment (17cm x 13cm). The unit is securely fastened to the servo horns, while an internal circuit of heating elements provides the necessary thermal control.

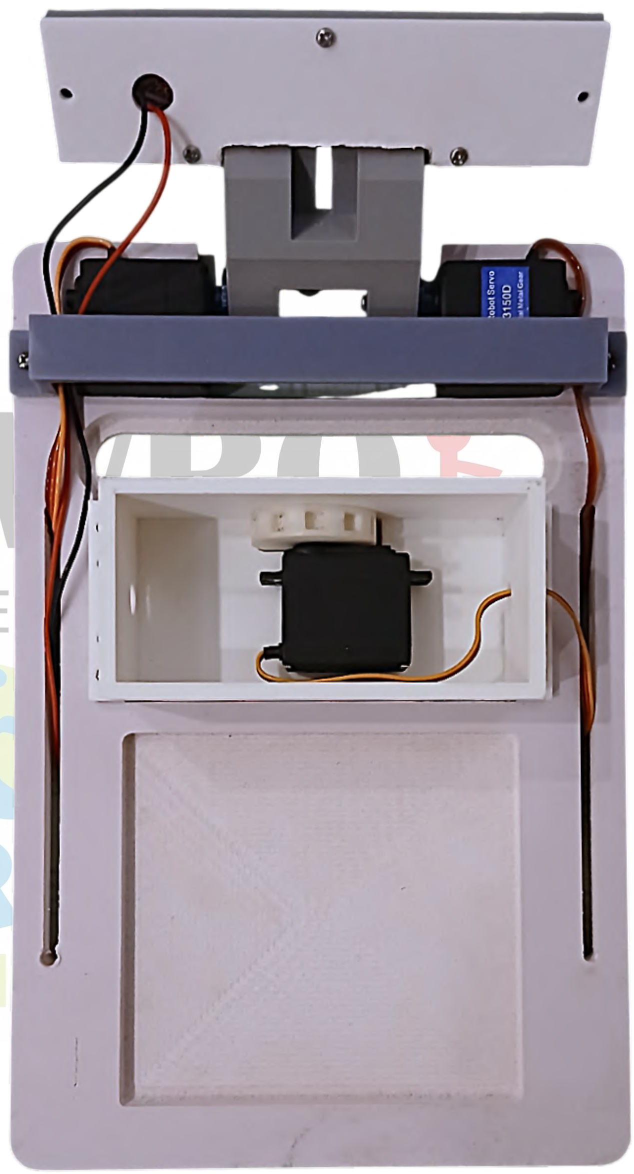

Figure 10: Integration of the seeding compartment onto the PVC baseplate (15cm x 6.5cm). The assembly is secured via screws and houses an internal servo motor that drives a cylindrical seeding head (4cm diameter x 1cm thickness). Grooves within the head capture seeds and transport them to the opening at the base, where they are released upon rotation.

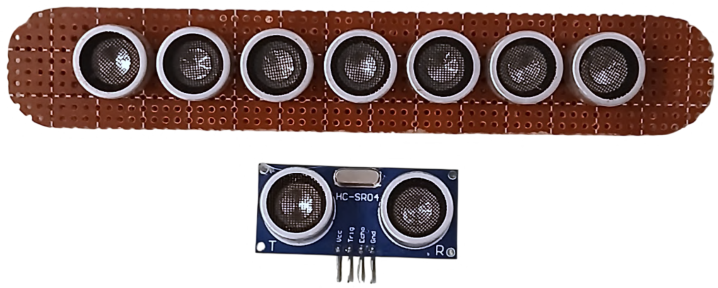

Figure 11: Ultrasound circuit wired in parallel (16cm x 3cm). An Arduino ultrasonic distance sensor is utilized for the ultrasound transducers.

Figure 12: Seeding holding plate installed above the seeding compartment. The plate acts as a reservoir for seeds and features a slight curvature designed to channel them toward the seeding head for distribution.

Figure 13: Ultrasound subsystem integrated onto the baseplate. The circuit is protected and securely fastened via a dedicated cover.

Figure 14: Watering compartment fastened to the baseplate (15cm x 12cm). The container is shaped to funnel water toward the pump intake, which feeds a rear-mounted sprinkler via a connecting pipe. The compartment includes side openings for wire and pipe routing.

Figure 15: Top cover installed and secured to the watering compartment to seal the main reservoir. Additionally, the side routing holes have been manually sealed to prevent any spillage during operation.

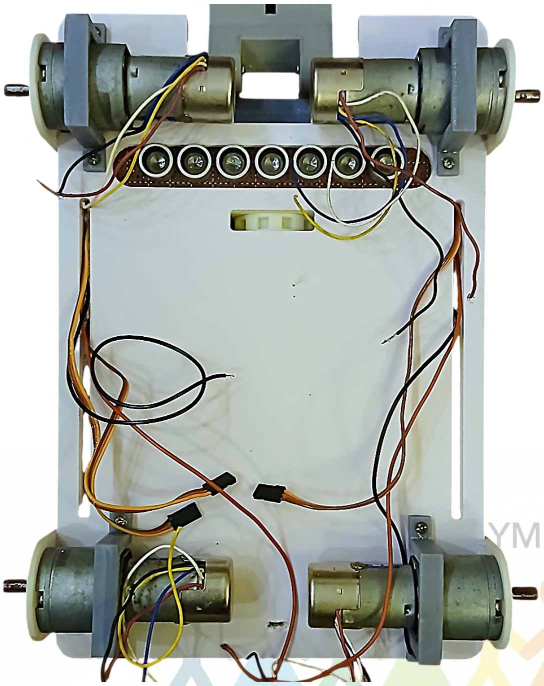

Figure 16: Encoder motors secured by two separate holders: one screwed into the side and one into the bottom. These provide high structural strength for rough terrain.

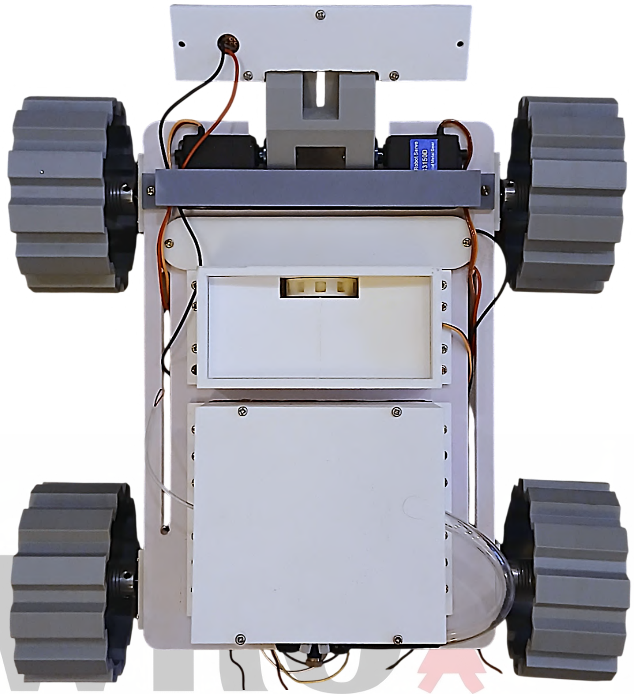

Figure 17: 3D-printed tires (10cm diameter x 4.5cm thickness) attached via motor couplers. The tires feature deep grooves to ensure traction in soil.

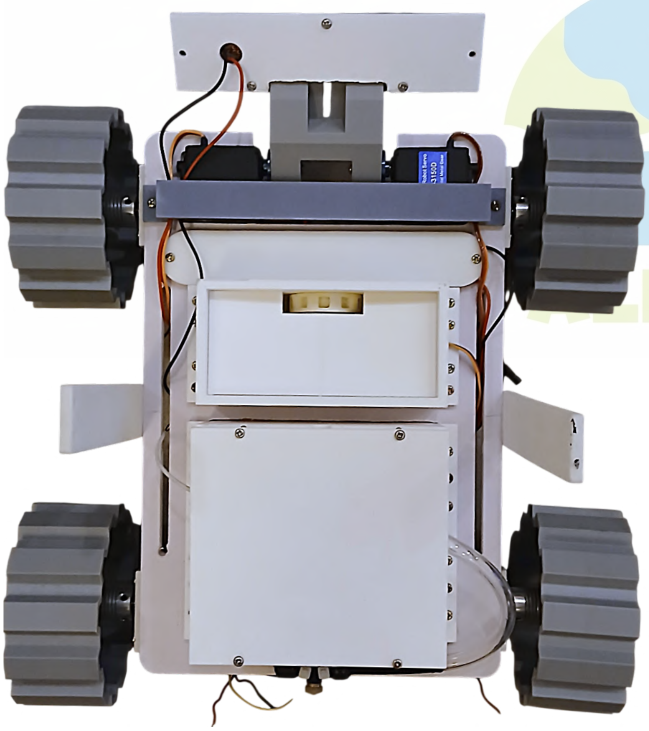

Figure 18: Dual vertical rods attached to the center of the baseplate sides (15cm x 3cm). These serve as the mounting frame for the solar power system.

Figure 19: Solar panel attached to its mounting plate, which is screwed securely into the vertical holders from both sides (20cm x 27cm).

V. Electrical Design

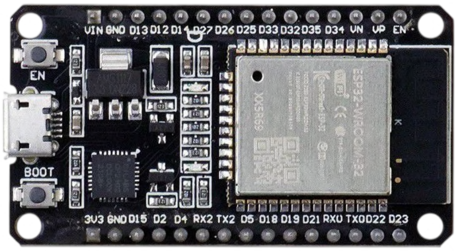

Figure 20: ESP32 microcontroller (”the brain”). Chosen for high clock speeds and processing power superior to standard Arduino boards.

Figure 21: TowerPro MG996R metal-gear servo. Drives the seeding head for reliable seed distribution.

Figure 22: HDKJ S3150D high-torque servo. Features 15 kg/cm stall torque and metal gears, providing the high force needed.

Figure 23: 6V DC water pump motor used to drive the irrigation system.

Figure 24: High-torque 24V motors (run at 12V) with a 224:1 gear reduction. Recycled units retrofitted with encoders.

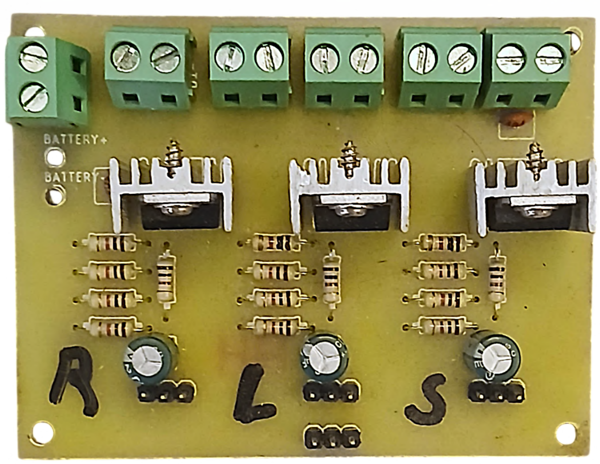

Figure 25: Custom PCB with voltage regulators and heatsinks. Manages high current for servos and links the battery, solar panel, and motor drivers.

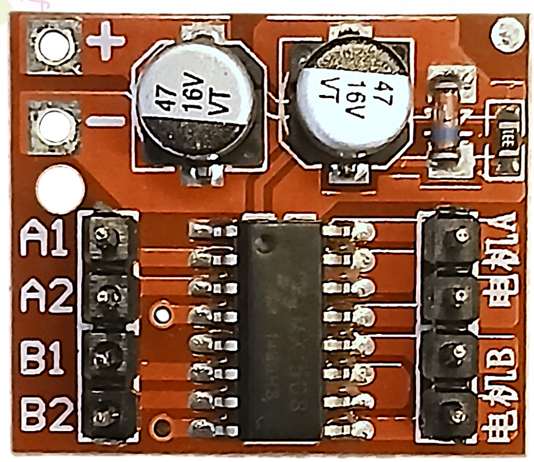

Figure 26: L298N Mini motor driver module. A compact dual-channel H-bridge used to control the speed and direction of the rover’s drive motors, alongside supplying power for the pump and heating mechanism.

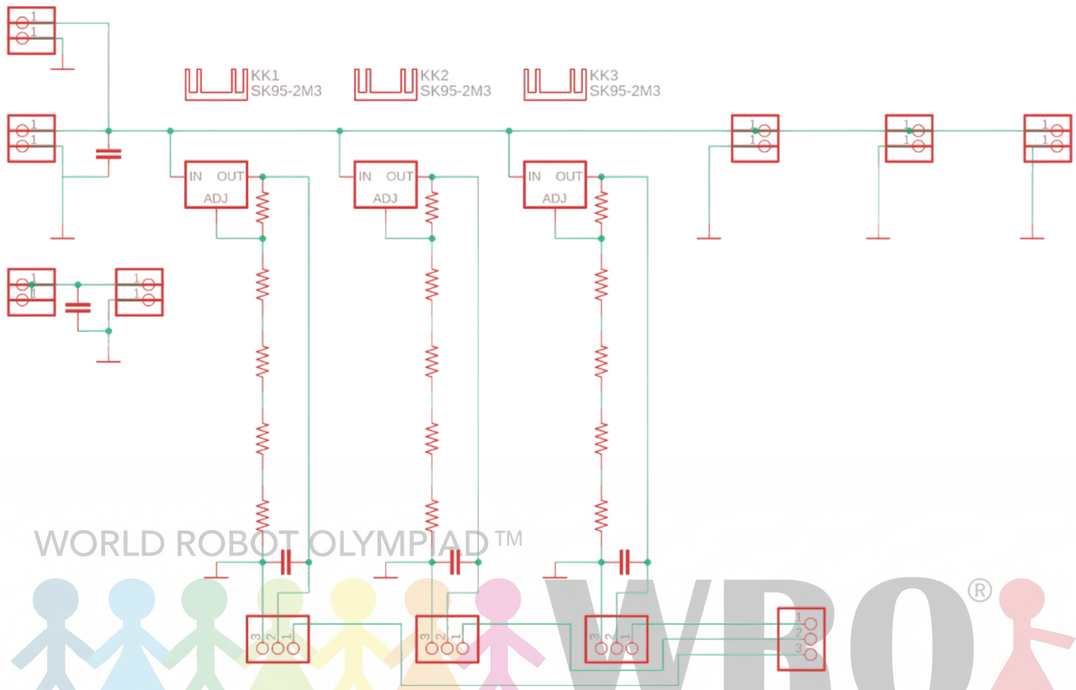

Figure 27: Complete circuit schematic designed in EAGLE CAD. It uses LM317T voltage regulators set to 6V, providing up to 1.5A of current for each servo. Heat sinks are included to manage thermal load. Includes pins for the 3 servos and additional pins for signal connections to the ESP32

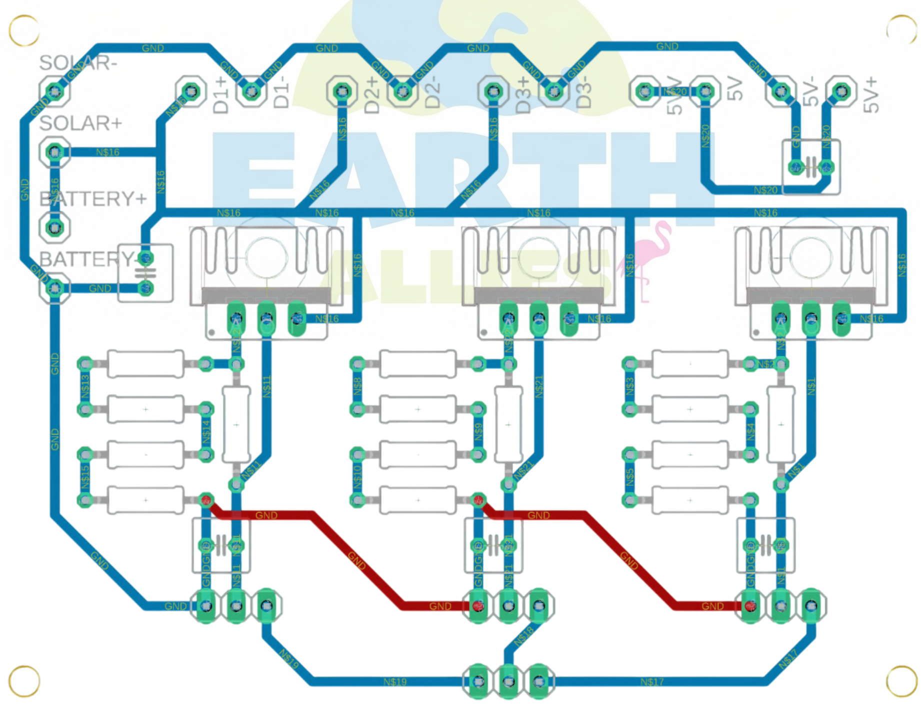

Figure 28: Single-layer PCB layout designed in EAGLE CAD. Due to space and routing constraints on a single layer, specific traces were replaced with manual wire jumpers to complete the circuit (red wires in the figure).

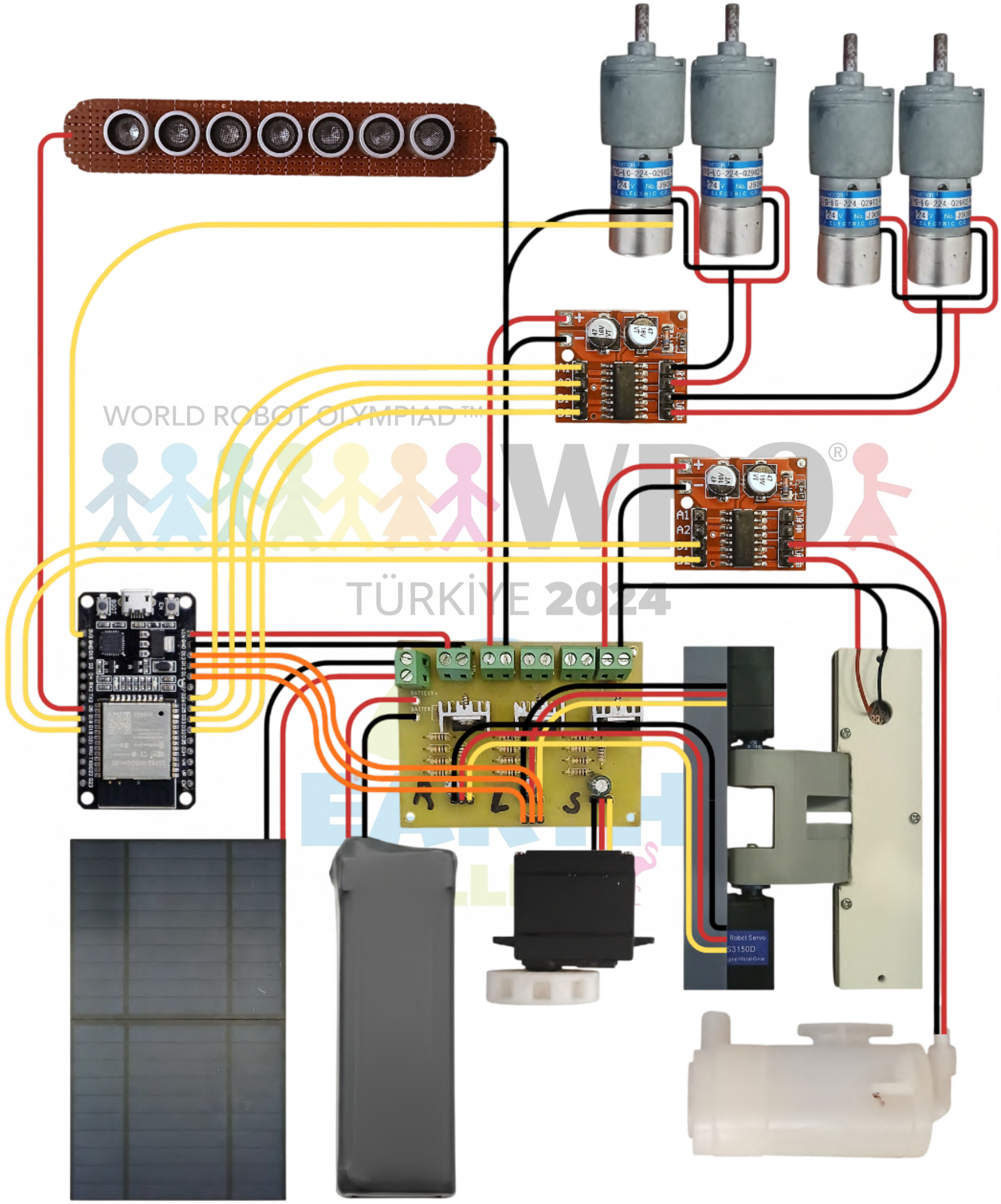

Figure 29: Final SonoSeeder circuit assembly. The four drive motors are wired in two parallel pairs (left and right) to be controlled by a single motor driver. A second driver manages the water pump and heating element. The system is powered by a 11.1V 2200mAh battery. The solar panels (6V each) are wired in series to provide a suitable 12V.

VI. Control and Software

This section reviews the key software components instrumental to the operation of SonoSeeder.

```
void setup() {
  Serial.begin(115200);
  SerialBT.begin("ESP32test");

  while (!SerialBT.available())
    ;
  SerialBT.println("Welcome to SonoSeeder");
  delay(1000);

  SerialBT.println("Please input the following operation parameters:");
  String temp = SerialBT.readString();

  SerialBT.println("Please enter field length /cm");
  while (!SerialBT.available())
    ;
  in_Distance_Y = SerialBT.readString().toInt();

  SerialBT.println("Please enter field width /cm");
  while (!SerialBT.available())
    ;
  in_Distance_X = SerialBT.readString().toInt();

  SerialBT.println("Please enter path divisions");
  while (!SerialBT.available())
    ;
  Iterations = SerialBT.readString().toInt();

  SerialBT.println("Please enter Ultrasound frequency /KHz");
  while (!SerialBT.available())
    ;
  Ultrasound_freq = SerialBT.readString().toInt();

  Seeding.write(Seeding_pos);

  delay(1000);

  Front_left.write(90);
  Front_right.write(90);

  delay(2000);

  Front_left.write(90 - 70);
  Front_right.write(90 + 70);

  Target = 10 - 1;  //err value
  Target_turn = 6 - 1;

  while (!compass.begin()) {
    Serial.println("Could not find a valid 5883 sensor, check wiring!");
    delay(500);
  }

  int yaw_offset = 0;
  for (int i = 0; i < 100; i++) {
    updateYaw();
    yaw_offset += yaw;
```

```
  }
  yaw_offset /= 100;
  yaw -= yaw_offset;
}
```

The system initialization protocol and setup logic. The process begins by establishing a `Bluetooth` connection to receive mission parameters, including field dimensions (`X` and `Y`), path subdivisions (`Iterations`), and the `Ultrasound_frequency`. The `Iterations` decides the number of rows on which the SonoSeeder will operate. The field preparation compartment is then deployed into the soil; the servos transition from a neutral 90° position to $90 \pm 70°$, a configuration that ensures the two mirrored, face-to-face servos rotate in opposite directions to move the field preparation compartment in tandem. Target velocities are set for straight traversal (`10 RPM`) and turning (`6 RPM`), with a `-1` offset applied to compensate for observed mechanical overshoots. Finally, the system calibrates the orientation datum by averaging 100 magnetometer readings to establish a stable `yaw_offset`, effectively zeroing the compass so that the rover's starting direction is defined as 0°.

```
pinMode(23, INPUT);
attachInterrupt(23, time_stamp, RISING);
```


Interrupt setup on `Pin 23` for a single motor encoder. Because the motors move in synchronization at equal speeds, a single encoder provides sufficient data for displacement tracking. The `attachInterrupt` function triggers the `time_stamp` routine on a `RISING` edge, allowing the system to calculate real-time `RPM` and distance. During turns, the rover rotates on its central axis, requiring no displacement tracking, which eliminates the need for additional encoders.

```
void IRAM_ATTR time_stamp() {
  elapsed_time = micros() - prev_time;
  prev_time = micros();
}
```

Interrupt function calculates the time elapsed between pulses using `micros()`. This `IRAM_ATTR` designated routine ensures minimal latency by running from internal RAM. The difference between the current time and `prev_time` is stored in `elapsed_time`, which is later used to determine the `RPM`, angular displacement, and the rover's total distance traveled.

```
ledcAttachChannel(in1, 2000, 8, 1);
ledcAttachChannel(in2, 2000, 8, 2);
ledcAttachChannel(in3, 2000, 8, 3);
ledcAttachChannel(in4, 2000, 8, 4);
ledcAttachChannel(Pump, 2000, 8, 5);
ledcAttachChannel(Heat, 2000, 8, 6);
ledcAttachChannel(Ultrasound, Ultrasound_freq * 1000, 8, 7);
ledcAttachChannel(Seeding_pin, 50, 10, 8);
ledcAttachChannel(Front_left_pin, 50, 10, 9);
ledcAttachChannel(Front_right_pin, 50, 10, 10);

Seeding.attach(Seeding_pin);
Front_left.attach(Front_left_pin);
Front_right.attach(Front_right_pin);
}
```

Pinout setup utilizing the `ledcAttachChannel` function to define `PWM` frequencies across various subsystems. The motors, heating, and pump frequencies are set to `2000Hz`, a low value chosen to minimize current spikes caused by the parasitic capacitance of the motor driver inputs. This function also sets the specific `Ultrasound_freq` . For high-precision servo components, the frequency is set to `50Hz` with a `10-bit` resolution to provide the standard signal timing required for servos.

```
void forward(int speed, int addon_val) {
  analogWrite(in1, speed);
  analogWrite(in2, 0);
  analogWrite(in3, speed + addon_val);
  analogWrite(in4, 0);
}

void turn(int speed) {
  if (speed > 0) {
    analogWrite(in1, speed);
    analogWrite(in2, 0);
    analogWrite(in3, 0);
    analogWrite(in4, speed);
  }
  if (speed < 0) {
    analogWrite(in1, 0);
    analogWrite(in2, abs(speed));
    analogWrite(in3, abs(speed));
    analogWrite(in4, 0);
  }
  if (speed == 0) {
    analogWrite(in1, 0);
    analogWrite(in2, 0);
    analogWrite(in3, 0);
    analogWrite(in4, 0);
  }
}
```

Low-level motor control functions for rover movement. The `forward` function utilizes an `addon_val` from the yaw controller to adjust the speed of one motor side, correcting steering drift in real-time. The `turn` function facilitates zero-radius rotation by driving the left and right motor pairs in opposite directions based on the signed speed input.

```
void updateYaw() {
  sVector_t mag = compass.readRaw();
  static float hi_cal[3];

  float mag_data[] = { mag.XAxis / 10,
                       mag.YAxis / 10,
                       mag.ZAxis / 10 };

  // Apply hard-iron offsets
  for (uint8_t i = 0; i < 3; i++) {
    hi_cal[i] = mag_data[i] - hard_iron[i];
  }

  // Apply soft-iron scaling
  for (uint8_t i = 0; i < 3; i++) {
    mag_data[i] = (soft_iron[i][0] * hi_cal[0]) + (soft_iron[i][1] * hi_cal[1]) +
    (soft_iron[i][2] * hi_cal[2]);
  }

  heading = atan2(mag_data[1], mag_data[0]) * 180 / M_PI;

  if (prev_heading) {
    if (prev_heading > 90 && prev_heading <= 180 && heading < -90 && heading >= -180) {
      yaw += (180 - prev_heading) + (180 + heading);
    } else if (heading > 90 && heading <= 180 && prev_heading < -90 && prev_heading >= -180) {
      yaw -= (180 + prev_heading) + (180 - heading);
```

```
    } else yaw += heading - prev_heading;
  }

  prev_heading = heading;
}
```

Yaw calculation and compass calibration logic. The `updateYaw` function applies `hard_iron` offsets and `soft_iron` scaling matrices to raw magnetometer data to correct for interference from the rover's chassis and externally. It calculates the magnetic `heading` using `atan2` and implements a wrapping algorithm to maintain a continuous `yaw` value. This algorithm detects when the sensor crosses the $\pm 180^\circ$ boundary, preventing sudden jumps in the data and ensuring that the rover can track cumulative rotation accurately across multiple full turns.

```
const float hard_iron[3] = {
  65.203435, 54.533258, 164.871260
};

// Soft-iron calibration settings
const float soft_iron[3][3] = {
  { 0.477467, -0.001965, -0.044405 },
  { -0.001965, 0.472935, -0.036471 },
  { -0.044405, -0.036471, 0.473685 }
};
```

Correction matrices for soft iron and hard iron interference. These values were calculated using the Magneto 1.2 application. These values may be subject to change depending on location, recalibration is required.

```
void Traverse_Y(int Distance, int Maintain) {
  while (Distance_Y < Distance) {
    if (elapsed_time) RPM = ((60000000 / ((double)elapsed_time * 224 * 2)));
    Serial.print(RPM);
    Serial.print("   ");
    Serial.print(yaw);
    Serial.println();

    if (d_prev_time) d_elapsed_time = micros() - d_prev_time;
    d_prev_time = micros();

    seeding_mechanism(175, 15);

    double speed = ((RPM + 1) * 2 * 5 * M_PI) / 60;
    Distance_Y += (d_elapsed_time * speed) / 1000000;

    double err = Target - RPM;
    PWM += 0.02 * err;

    tracking = 10 * (yaw - Maintain);

    forward(PWM + base_pwm, tracking);
    updateYaw();
  }
  turn(0);
  Distance_Y = 0;
  Distance_X = 0;
  elapsed_time = 0;
  d_elapsed_time = 0;
  d_prev_time = 0;
```

```
  s_elapsed_time = 0;
  PWM = 0;
  turn_PWM = 0;
  tracking = 0;
}
```

Field traversal logic for the `Y-axis`. This function uses encoder data to calculate real-time `RPM` and linear speed, integrating them to track total displacement (`Distance_Y`). The calculation incorporates a gear reduction ratio of `224:1` and dual-edge triggering (`* 2`) to derive accurate `RPM` from the `elapsed_time` between pulses. It implements a `P-controller` to maintain the `Target` speed by dynamically adjusting the `PWM` output based on the velocity error. To determine displacement, the function calculates linear velocity using the circumference of the tire ($2 \times 5 \times \pi$) and integrates this over `d_elapsed_time`. Simultaneously, it utilizes `yaw` feedback to calculate a `tracking` value, correcting any steering drift relative to the `Maintain` angle. The `seeding_mechanism` is integrated directly into this loop for automated dispersal during movement. An identical function, `Traverse_X`, handles horizontal shifts. Upon completion, all navigation variables, including `Distance_Y` and `PWM`, are reset to ensure zero-referenced accuracy for the subsequent maneuver.

```
void seeding_mechanism(int angle, int time) {
  s_elapsed_time += d_elapsed_time;

  if (s_elapsed_time >= time * 1000) {
    s_elapsed_time = 0;

    if (Seeding_pos == angle) {
      state = 1;
    }

    if (Seeding_pos == 0) {
      state = 0;
    }

    if (state == 0) Seeding_pos++;
    else Seeding_pos--;

    Seeding.write(Seeding_pos);
  }
}
```

The periodic seeding mechanism control logic. This function utilizes a non-blocking timing approach by accumulating `d_elapsed_time` and comparing it against a target threshold, rather than using a standard `delay()`. By employing conditional `if` statements to increment or decrement the `Seeding_pos` one degree at a time, the function allows the seeding servo to oscillate between 0° and the specified `angle` at a controlled speed. This design ensures that the seeding operation can execute concurrently within the `Traverse_Y` loops without interrupting time-sensitive calculations or sensor updates.

```
void clockwise_turn(int target_angle) {
  while (angle_err > 0) {
    if (elapsed_time) RPM = ((60000000 / ((double)elapsed_time * 224 * 2)));
    angle_err = (target_angle - yaw);

    double turn_err = Target_turn - RPM;
    turn_PWM += 0.02 * turn_err;

    turn(base_pwm + turn_PWM);
    updateYaw();
  }
  Distance_Y = 0;
  Distance_X = 0;
```

```
  elapsed_time = 0;
  d_elapsed_time = 0;
  d_prev_time = 0;
  s_elapsed_time = 0;
  PWM = 0;
  turn_PWM = 0;
  tracking = 0;
  angle_err = 1;
}
```

Rotation control logic for orientation adjustments. This function executes a controlled clockwise turn by monitoring the `angle_err` between the current `yaw` and the `target_angle`. An identical function exists for anti-clockwise rotations. Both utilize closed-loop `RPM` feedback using a `P-controller` to maintain a consistent turning velocity via the `turn_PWM` variable and reset navigation parameters, such as `Distance_Y` and `tracking`, upon completion to ensure precise alignment for the next traversal segment.

```
void loop() {
  analogWrite(Heat, 100);
  for (int n = 0; n < Iterations; n++) {
    analogWrite(Pump, 100);
    Traverse_Y(in_Distance_Y, 0);

    Front_left.write(90);
    Front_right.write(90);
    analogWrite(Pump, 0);

    clockwise_turn(90);
    Traverse_X(in_Distance_X / (2 * Iterations), 90);
    clockwise_turn(180);

    Front_left.write(90 - 70);
    Front_right.write(90 + 70);

    analogWrite(Pump, 100);
    Traverse_Y(in_Distance_Y, 180);

    Front_left.write(90);
    Front_right.write(90);
    analogWrite(Pump, 0);

    anticlockwise_turn(90);
    Traverse_X(in_Distance_X / (2 * Iterations), 90);
    anticlockwise_turn(0);

    Front_left.write(90 - 70);
    Front_right.write(90 + 70);
  }

  Front_left.write(90);
  Front_right.write(90);

  while (1) {
    turn(0);
    Serial.print("done");
  }
  analogWrite(Heat, 0);
}
```

The complete autonomous operation loop. The process begins by activating the `Heat` element, which remains active throughout the mission. The irrigation `Pump` is dynamically toggled, activating specifically during the `Traverse_Y` function to provide water only during linear row coverage. Integrated within this same traversal function is the `seeding_mechanism`. For field preparation, the field preparation compartment is lowered into the soil during Y-axis movement by writing the mirrored 90 ± 70 values to the `Front_left` and `Front_right` servos. The system automatically raises the field preparation compartment to `90` degrees (deactivated) during turning maneuvers; this is essential because the mechanical resistance of the compartment in the soil would prevent the rover from turning. Upon completion of the turn and the `Traverse_X` lateral shift, the compartment is redeployed for the next row. Once the specified `Iterations` are complete, the system halts movement and signals "done" via the serial monitor.

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